Javad Omidi

Postdoctoral Associate
Cornell University, Ithaca, NY, US
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U.S. Permanent Resident (Green Card Holder)

Professional Summary

Translational and computational oncology researcher with 14+ years of quantitative and computational experience spanning algorithmic modeling, optimization, engineering computation, and data-intensive research, including 6+ years of independent research in cancer genomics, bioinformatics, systems biology, and precision oncology. Integrates experimental molecular biology with computational genomics, with hands-on experience in DNA/RNA extraction, PCR/qPCR, digital PCR (dPCR), reverse transcription, flow cytometry, in vitro cell culture, and whole-genome/whole-exome sequencing preparation. Research expertise includes WGS/WES, bulk and spatial transcriptomics, RNA-seq/miRNA-seq, cfDNA genomics, somatic variants and CNV analysis, multi-omics integration, and regulatory-network reconstruction using Python/R and HPC workflows. Leads end-to-end biomarker and therapeutic target discovery using machine learning, deep learning, survival modeling, molecular docking, optimization algorithms, and robust cross-validated inference. Current research focuses on comparative and precision oncology, including liquid-biopsy genomics, tumor heterogeneity, cross-species molecular conservation, spatially resolved transcriptomic profiling, miRNA biomarker prioritization, and molecular biomarker discovery in canine and feline lymphoma with translational relevance to human cancer.

Academics

Postdoctoral Associate — Mar 2026 – Mar 2027
Cornell University, Ithaca, NY, US
Department of Population Medicine & Diagnostic Sciences
Clinical Pathology, Cytopathology Research Lab

PhD, "Chemical Engineering" — Jan 2022 – Feb 2026
Columbia University, New York City, NY, US
CGPA: 4.0/4.0

MPhil, "Chemical Engineering" — Jan 2022 – Dec 2023
Columbia University, New York City, NY, US
CGPA: 4.0/4.0

MS, "Engineering" — Sep 2012 – Jan 2015
Sharif University of Technology, Tehran, Iran
CGPA: 3.8/4.0

BS, "Engineering" — Sep 2007 – Sep 2012
Amirkabir University of Technology (Tehran Polytechnic), Tehran, Iran
CGPA: 3.2/4.0

High School & Pre-University, "Mathematics and Physics" — Sep 2003 – Sep 2007
Farhangian High School, Arak, Iran
CGPA: 4.0/4.0

Research Projects

Cross-Species Discovery of Conserved Molecular Networks in Canine and Human B-Cell Lymphoma
Supervisor: Prof. Priscila Serpa, Cornell University, Ithaca, NY, US — Mar 2026 – Now

- 103 plasma–tumor shared mutated genes identified
- 95.35% of canine orthologs represented in human lymphoma
- 13 conserved cancer drivers, including MYC, MTOR, ARID1A, and POT1
- 21 conserved pathways across canine and human lymphoma
- RHO GTPase, PI3K–Akt, and cytoskeletal signaling strongly conserved
- MYC identified as the dominant protein-network hub

Genome-Wide cfDNA Profiling of Tumor Heterogeneity in Canine B-Cell Lymphoma
Supervisor: Prof. Priscila Serpa, Cornell University, Ithaca, NY, US — Mar 2026 – Now

- Matched WGS profiling of plasma cfDNA, tumor, and germline DNA
- Tumor–plasma variant concordance ranging from 1.7–49.3% across cases
- 174 shared mutated genes with substantial compartment-specific heterogeneity
- 712 CNVs revealing distinct plasma and tumor genomic architectures
- Conserved PI3K-Akt, JAK-STAT, integrin, and oncogenic signaling pathways
- Plasma-specific genomic signals supporting cfDNA-based liquid biopsy

Machine Learning–Based miRNA Biomarker Prioritization in Feline Intestinal Lymphoma
Supervisor: Prof. Priscila Serpa, Cornell University, Ithaca, NY, US Mar 2026 – Now

- DEG-based ranking of candidate miRNAs
- Machine learning–based LGITL–IBD discrimination
- Cross-validated miRNA feature prioritization
- Selection of top candidates for qPCR validation
- Novel miRNA annotation and homology assessment

Spatial Transcriptomics of Feline Intestinal Lymphoma and IBD
Supervisor: Prof. Priscila Serpa, Cornell University, Ithaca, NY, US Mar 2026 – Now

- Spatial transcriptomic profiling of feline LGITL and IBD
- Cell- and tissue-specific differential gene expression
- LGITL–IBD signatures localized to CD3+ T cells
- FYN, BCL2, PIK3CD, and DOCK2 as lymphoma-associated candidates
- Candidate molecular biomarkers for LGITL–IBD differentiation

Deep Learning–Based Cross-Disease Transcriptomic Integration of ACC and Major Depressive Disorder
Columbia University, New York, NY, US Mar 2025 – Feb 2026

- 30 shared dysregulated genes across ACC and MDD
- HPA-axis-linked inflammatory and immune signatures
- MMP9, LCN2, and PTX3 as central network mediators
- Joint autoencoder for cross-disease latent representation
- Shared molecular clusters transcending disease labels

AgomiR-466 Therapeutic Targeting of Oncogenic Pathways in ACC
Columbia University, New York, NY, US Nov 2024 – Oct 2025

- miR-466 downregulation in ACC, stage-specific loss
- Survival advantage with high miR-466 expression
- 394 upregulated oncogenic transcripts as miR-466 targets
- CHEK1, BUB1, MELK prioritized oncogenic drivers
- Stable RNA–RNA duplexes with AgomiR-466 binding
- Favorable HADDOCK docking scores and hydrogen bonding
- ADMET: safe profile, poor intestinal absorption limits

Machine Learning–Optimized Blood Biomarker Panel for MDD & OCD Diagnosis
Columbia University, New York, NY, US May 2024 – Oct 2025

- secretory blood biomarkers prioritized for MDD & OCD
- Commercial ELISA kits available for all candidates
- Immune-inflammatory pathways enriched in selected genes
- Network analysis highlights LTF, MMP8, LCN2 hubs
- Logistic regression best classifier for diagnostic stability
- Strong AUC/AUPRC in internal and external validation
- Identification of Novel biomarkers PF4V1, FOLR3, GZMK

Machine Learning Framework for ceRNA Network Analysis in Adrenocortical Carcinoma
Columbia University, New York, NY, US Sep 2023 – Sep 2024

- Random Forest highest accuracy in ceRNA interaction prediction
- Tumor network rewiring and reduced hub connectivity observed
- Novel miRNA–mRNA interactions in oncogenic signaling pathways
- Interaction validation through TargetScan and miRTarBase databases
- Normal ceRNA networks denser and more structurally robust
- Candidate biomarkers and therapeutic targets for adrenocortical carcinoma

Discovery of Central microRNAs via Systems Biology and Multi-Omics Analysis in Adrenocortical Carcinoma
Columbia University, New York, NY, US Oct 2022 – Dec 2024

- Extensive ceRNA network rewiring between normal and tumor tissues
- Central hub miRNAs regulate oncogenic and tumor-suppressive pathways
- Stage-specific expression dynamics reveal diagnostic and prognostic potential
- miRNA–mRNA inverse correlations confirm regulation of key oncogenes
- Therapy-responsive microRNAs identified with radiotherapy-sensitive expression changes
- Candidate biomarkers proposed for prognosis and therapeutic targeting in ACC

Novel Systems-Level ceRNA Network Framework for Oncogenic Biomarker Discovery in Adrenocortical Carcinoma
Columbia University, New York, NY, US Nov 2022 – Sep 2023

- Construction of tumor-specific ceRNA regulatory network
- Identification of hub miRNAs: miR-466, miR-940, miR-507
- Integrated miRNA–mRNA filtering and inverse expression validation
- Prioritization of four novel oncogenic biomarkers: CKS2, ERP44, ERG28, FAM32A
- Multivariate survival and ROC analyses confirming clinical utility
- Functional annotation revealing roles in cell cycle, apoptosis, ER stress

Research, Teaching, & Advising Experiences

Lab Manager May 2026 – Now
Cytopathology Research Lab, Dep. of Population Medicine & Diagnostic Sciences, Cornell University, Ithaca, NY, US
Managing laboratory operations, including oversight and maintenance of the lab website, coordination and organization of laboratory equipment and resources, and support for experimental activities. Assisting lab members with project planning, development, and management, and providing guidance on experimental and research workflows.

Postdoctoral Associate Mar 2026 – Now
Department of Population Medicine & Diagnostic Sciences, Cornell University, Ithaca, NY, US
Comparative Oncology • Precision Oncology • Computational Biology
Conducting translational and comparative oncology research integrating wet-lab molecular techniques with computational genomics and bioinformatics. Experimental work includes DNA/RNA extraction, PCR, qPCR, reverse transcription, flow cytometry, and preparation of whole-genome and whole-exome sequencing libraries. Computational research encompasses WGS/WES, RNA-seq/miRNA-seq, and spatial transcriptomics analysis; somatic variant and CNV characterization; differential expression and cell-/tissue-specific transcriptomic profiling; biomarker discovery; machine learning; and cross-species molecular analyses, with applications in canine and feline lymphoma and comparative human cancer research.

STEM Research Mentor Jul 2026 – Aug 2026
Herricks High School, New Hyde Park, NY, US
Mentored a high school student in cancer genomics and bioinformatics, guiding an independent research project comparing differential gene expression between adrenocortical carcinoma (ACC) and kidney renal clear cell carcinoma (KIRC) to identify shared candidate genes. Provided training in scientific literature review, research methodology, Python-based genomic data analysis, statistical analysis, and scientific writing.

Scientific Peer-Reviewed Journals

A) Scientific Peer Reviewer (Biology & Medicine) Mar 2022 – Now
Discover Oncology (Springer) • Immunobiology (Elsevier) • Advances in Cancer Biology - Metastasis (Elsevier) • Molecular and Cellular Biochemistry (Springer) • Computers in Biology and Medicine (Elsevier) • JMIR Research Protocols (JMIR) • JMIR Cancer (JMIR) • JMIRx Bio (JMIR) • JMIR Medical Informatics (JMIR) • Journal of Medical Internet Research (JMIR) • Biomarkers (Taylor & Francis) • Heliyon (Cell Press) • European Journal of Medical Research (Springer Nature) • Scientific Reports (Springer Nature) • Integrative Biology (Oxford University Press) • Oncology Research (Tech Science Press) • Advances in Biomarker Sciences and Technology (Elsevier) • Journal of Biomedical Research (Elsevier) • Gene Reports (Elsevier) • Cancer Informatics (Sage) • Oncologie (Tech Science Press) • Journal of Pure and Applied Microbiology (JPAM).

B) Scientific Peer Reviewer (Physical Sciences & Engineering) Oct 2016 – Now
AIAA Journal (American Institute of Aeronautics and Astronautics) • Journal of Applied Physics (American Institute of Physics) • Physics of Fluids (American Institute of Physics) • Chemical Engineering Science (Elsevier) • Chemical Engineering Journal (Elsevier) • Powder Technology (Elsevier) • IEEE Access (Institute of Electrical and Electronics Engineers) • Heliyon (Cell Press) • Scientific Reports (Springer Nature) • Energy Engineering (Tech Science Press) • Wind Engineering (Sage).

Research Assistant (RA) Jan 2022 – Dec 2025
Department of Chemical Engineering, Columbia University, New York City, NY, US
Boyce Lab, PI: Prof. Chris Boyce
Developed computational pipelines (Python/R) for large-scale data analysis; experience with high-throughput data processing (Linux/HPC); collaborated with interdisciplinary teams of engineers and data scientists

Independent Researcher Jan 2022 – Dec 2025
Department of Chemical Engineering, Columbia University, New York City, NY, US
Led independent multi-omics and cancer genomics projects involving bulk RNA-seq, miRNA-seq pipeline development, ceRNA network analysis, biomarker discovery, and regulatory network modeling in cancer / using AI for biomarker discovery in rare cancers and psychiatric disorders

Teaching Assistant (TA) Jan 2022 – May 2023
Department of Chemical Engineering, Columbia University, New York City, NY, US

- Graduate Course, Transport in Fluid Mixtures, Prof. Chris Durning — Spring 2023
- Graduate Course, Math Methods in Chemical Engineering, Prof. Venkat Venkatasubramanian — Fall 2022
- Graduate Course, Advanced Chemical Kinetics, Dr. Robert Bozic — Spring 2022

Delivered computational-modeling modules and supervised project-based learning for graduate cohorts in fluid transport, kinetics, and mathematical modeling, enhancing course engagement metrics

Workshop Organizer Team Jan 2023
18th Northeast Complex Fluids and Soft Matter Workshop
Department of Chemical Engineering, Columbia University, NY, US

Coordinated logistics, scientific programming, and panel sessions for 200+ attendees, fostering cross-institutional collaboration in soft-matter and granular-materials research

Research Assistant (RA) Feb 2015 – Dec 2021
Sharif University of Technology, Tehran, Iran
PI: Prof. Karim Mazaheri

Developed advanced numerical & algorithmic frameworks for plasma–flow modeling, enabling high-fidelity large-scale simulations and resulting in five Q1 publications and a Nature-indexed article.

Teaching Assistant (TA) Feb 2014 – Jan 2015
Sharif University of Technology, Tehran, Iran

- Undergraduate Course, Aerodynamics I — Fall 2014
- Graduate Course, CFD I — Fall 2014
- Graduate Course, Wind Turbine Aerodynamics — Fall 2013
- Graduate Course, Aerodynamics II — Fall 2013
- Graduate Course, CFD I — Fall 2013
- Undergraduate Course, Heat Transfer — Spring 2013

Thesis Advisor Dec 2015 – Jan 2017
Sharif University of Technology, Tehran, Iran

- **MS Thesis Title:** *Numerical Simulation of DBD Plasma Actuator and Optimization with Differential Evolution (DE) Algorithm for Separation Control,* **By:** Sajjad Jafari, **Supervisor:** Prof. Karim Mazaheri
- **MS Thesis Title:** *Aerodynamic Performance Improvement of Megawatt Wind Turbine Rotor Using Plasma Actuator,* **By:** Mohammadreza Movahhedi, **Supervisor:** Dr. Abbas Ebrahimi

Designed and graded advanced math and computational coursework for over 150 engineering students, integrating simulation-based assignments and case studies

Training Workshop on C++ Programming & Application in CFD Oct 2014
Sharif University of Technology, Tehran, Iran

Educational/Emotional Counseling and Academic Planning for University Entrance Exam Applicants
Arak Library, Arak City, Iran Sep 2014 – Sep 2015

Journal Publications

Immune, metabolic, and steroidogenesis-associated profiles of miR-940, miR-375, and miR-326 in adrenocortical carcinoma
Integrative Biology (doi.org/10.1093/intbio/zyag016) 2026 (Published)
Javad Omidi

AgomiR-based Therapeutic Targeting of Oncogenic Pathways Regulated by miR-466 in Adrenocortical Carcinoma
Journal of Pharmacology and Pharmacotherapeutics (doi.org/10.1177/0976500x261462664) 2026 (Published)
Javad Omidi

Integrative machine learning-driven prioritization of ceRNA networks in adrenocortical carcinoma
Intelligent Oncology (doi.org/10.1016/j.intonc.2026.100056) 2026 (Published)
Javad Omidi

miR-466 as a central miRNA with tumor-suppressive potential in the regulatory network of adrenocortical carcinoma
In Silico Research in Biomedicine (doi.org/10.1016/j.insi.2026.100214) 2026 (Published)
Javad Omidi

miR-507 and miR-665 as central MicroRNA regulators in the ceRNA network of adrenocortical carcinoma: A systems biology approach
Human Gene (doi.org/10.1016/j.humgen.2025.201498) 2025 (Published)
Javad Omidi

Synergistic dual oncogenic role of CKS2 and ACAT2: Enhanced regulatory coherence and biomarker potential in adrenocortical carcinoma
Cancer Genetics (doi.org/10.1016/j.cancergen.2025.10.103) 2025 (Published)
Javad Omidi

Molecular Landscape and Biomarker Discovery in Adrenocortical Carcinoma: An Integrative Review of Bioinformatics and Translational Insights
Pathology - Research and Practice (doi.org/10.1016/j.prp.2025.156295) 2025 (Published)
Javad Omidi

Therapy-Associated Oncogenic Axes of Core Cellular and Metabolic Programs Revealed by Multi-Omics Regulatory Network Analysis in Adrenocortical Carcinoma
medRxiv (doi.org/10.64898/2025.12.21.25342239) 2025 (Preprint)
Javad Omidi

Cross-Species Molecular Conservation of RHO GTPase, PI3K–Akt, and Cytoskeletal Signaling in Canine and Human B-Cell Lymphoma
Javad Omidi, Sai Navya Vadlamudi, Priscila B. S. Serpa 2026 (Ready to Submit)

Genome-Wide Characterization of Tumor-Derived Cell-Free DNA in Canine B-Cell Lymphoma Using Matched Plasma-Tumor Sequencing
Javad Omidi, Sai Navya Vadlamudi, Priscila B. S. Serpa 2026 (Ready to Submit)

Spatial Transcriptomic Analysis of Low-Grade Intestinal T-Cell Lymphoma and Lymphoplasmacytic Inflammatory Bowel Disease in Cats
Laura Victoria Quishpe Contreras, Javad Omidi, Priscila B. S. Serpa 2026 (Ready to Submit)

Peripheral Blood Secretome Biomarkers for Obsessive-Compulsive Disorder: A Machine Learning-Driven Diagnostic Panel
Frontiers in Genetics 2026 (Under Review)
Javad Omidi, Niloufar Salimian

HPA-Axis-Linked Inflammatory Crosstalk Between Adrenocortical Carcinoma and Major Depressive Disorder in a Joint Autoencoder Latent Space
Advances in Biomarker Sciences and Technology 2026 (Under Review)
Javad Omidi, Niloufar Salimian

For the complete list of publications, please see my **Google Scholar**.

Invited Publications

Invited Book Chapter – Multi-Omics Networks in Adrenocortical Carcinoma: Biomarkers and Therapeutic Targets
Advances in Clinical Chemistry 2026 (Ready to Submit)

Invited Editorial – From Benchmark Performance to Clinical Trust: What Is Needed for LLM-Assisted Radiology Reporting?
JMIR Medical Informatics 2026 (Ready to Submit)

Conference Presentations

Can Dogs Decode Conserved Molecular Networks in Human B-Cell Lymphoma?
BMCB-GGD Symposium, Ithaca, New York, US 2026, Oral Presentation
Javad Omidi, Sai Navya Vadlamudi, Priscila B. S. Serpa

Cross-Species Discovery of Conserved Molecular Networks in Canine and Human B-Cell Lymphoma
25th Annual Biomedical & Biological Science (BBS) Symposium, Ithaca, New York, US 2026, Oral Presentation
Javad Omidi, Sai Navya Vadlamudi, Priscila B. S. Serpa

Comparative cfDNA-Tumor Profiling of Genomic Heterogeneity in Canine B-Cell Lymphoma
ASHG Annual Meeting, Montreal, Quebec, Canada 2026, Poster Presentation
Javad Omidi, Sai Navya Vadlamudi, Priscila B. S. Serpa

Circulating Tumor DNA Captures a Broader Mutational Spectrum and Core Oncogenic Signaling in Canine B-Cell Lymphoma Compared to Single-Site Tumor Biopsies
American College of Veterinary Pathologists Annual Meeting, Indianapolis, Indiana, US 2026, Oral Presentation
Priscila B. S. Serpa, Javad Omidi, Sai Navya Vadlamudi, Shawna Klahn

Proof-of-concept study of microRNA expression in intestinal samples from cats with low-grade intestinal lymphoma and lymphoplasmacytic inflammatory bowel disease
American College of Veterinary Pathologists Annual Meeting, Indianapolis, Indiana, US 2026, Oral Presentation
Laura V. Quishpe Contreras, Javad Omidi, Priscila B. S. Serpa

Identification of Key Oncogenic Genes and Tumor Suppressive miRNAs in Adrenocortical Carcinoma via ceRNA Network Analysis
ASHG Annual Meeting, Boston, Massachusetts, US 2025, Poster Presentation
Javad Omidi

Computer & Programming Skills

Programming: C++ • FORTRAN • MATLAB • Python / R Programming (RStudio) • Maple • EES (Engineering Equation Solver).

Software: COMSOL Multiphysics • ANSYS Fluent • OpenFOAM • GAMBIT • AutoCAD • CATIA • IBM SPSS Statistics • ParaView • Notebook/JupyterLab • Linux/Unix • 3D Max • SolidWorks • Photoshop • Microsoft Office • Tecplot • MFiX Multiphase Flow Simulation • USCF ChimeraX • LigPlot+ • HADDOCK 2.4 • Git/GitHub.

Professional Skills

Molecular Biology & Laboratory Skills: DNA & RNA Extraction and Purification • PCR & Quantitative PCR (qPCR) • Digital PCR (dPCR) • Reverse Transcription (RT) & cDNA Preparation • Flow Cytometry • Whole-Genome Sequencing (WGS) Library Preparation • Whole-Exome Sequencing (WES) Library Preparation • DNA/RNA Quantification & Quality Assessment (NanoDrop, Qubit) • Sample Preparation for Genomic & Transcriptomic Analysis • Molecular Assay Preparation & Experimental Validation • In Vitro Cell Culture • Aseptic Cell Culture Techniques • Cell Passaging & Subculturing • Cell Cryopreservation & Freezing • Cell Thawing & Recovery • Cell Counting & Viability Assessment • MRI & Optical Imaging • Image Analysis.

Project Management & Leadership: Research Project Design • Grant & Proposal Writing • Academic Advising • Workshop & Training Program Organization • Thesis Supervision & Research Topic Definition.

Teaching, Mentorship & Academic Service: Teaching Assistance in Graduate & Undergraduate Courses • Curriculum & Content Development • Mentorship & Student Advising • Peer Review for International Known Journals • Conference Presentation & Workshop Facilitation.

Research & Analytical Skills: Experimental Design & Validation • Multi-Omics Data Integration & Systems Biology • Whole-Genome & Whole-Exome Sequencing (WGS/WES) • RNA-seq & miRNA-seq Analysis • Spatial Transcriptomics • Somatic Variant Calling, Annotation & Prioritization • Copy Number Variation (CNV) Analysis • cfDNA & Liquid Biopsy Genomics • Differential Gene & miRNA Expression Analysis • Novel miRNA Discovery, Annotation & Homology Analysis • Comparative Genomics & Cross-Species Ortholog Analysis • Statistical Genomics • Sequencing Data Quality Control & Bioinformatics Pipeline Development • ceRNA & Regulatory Network Analysis • Protein–Protein Interaction & Molecular Network Analysis • Cancer Driver Gene & Pathway Analysis • Biomarker Discovery, Prioritization & Validation • Survival & Prognostic Modeling • Statistical Modeling & Hypothesis Testing • Network Pharmacology & Systems-Level Pathway Analysis • Structural Bioinformatics & Molecular Docking • Translational & Comparative Oncology • Computational Modeling & Simulation • Multi-Physics & Multi-Scale Modeling & Simulation • Electrohydrodynamics (EHD) & Magnetohydrodynamics (MHD) Modeling • Advanced Numerical Methods & Optimization Algorithms • Magnetic Resonance Imaging (MRI) Data Acquisition & Analysis • Optical & High-Speed Imaging • Scientific Visualization • Literature Review & Critical Analysis.

Data Science & Bioinformatics Skills: Genomic & Transcriptomic Databases/Resources: (TCGA • GTEx • GEO • COSMIC • mirBase • STRING • DAVID • IntOGen • ENCORI StarBase • Biomart • TCGA-ACC • GTEx • miRNATissueAtlas • miRTarBase • TargetScan • STRING • DAVID • Ensembl • miRBase • ENCORI (starBase) • IntaRNA • RNAhybrid • RNAfold (ViennaRNA) • RNA Composer • DuplexFold (RNAstructure) • NCI/CADD Chemical Identifier Resolver • pkCSM • SEA (Similarity Ensemble Approach) • ADMETlab 3.0 • ProTox-II • SRplot • SwissTargetPrediction • DIANA • GeneCard • UniProt • g:profiler • oncomiR • DGIdb • ctdBase • PHAROS • Gepia2 • ProteinAtlas • Expression Atlas • miRGator • RNA Central • OncoKB • ONGene • TSGene); Workflow Automation; Machine Learning for Bioinformatics; Deep Learning for Medicine; Data Preprocessing, Feature Engineering & Cross-validation; Cloud-Based Bioinformatics; Clinical Data Integration & Translational Bioinformatics; Immunoinformatics & Vaccine Design; AI for Medical Diagnosis, Prognosis & Treatment; Feature Selection & Dimensionality Reduction; WGS/WES Bioinformatics; Variant Calling, Filtering & Annotation; VCF Data Processing & Analysis; CNV Analysis; RNA-seq/miRNA-seq Differential Expression Analysis; Spatial Transcriptomics Data Analysis; Comparative & Cross-Species Genomics; Ortholog Mapping & Conservation Analysis; Pathway & Functional Enrichment Analysis; Reproducible Bioinformatics Pipeline Development; High-Performance Computing (HPC).

Communication & Professional Skills: Scientific Writing & Publication in High-Impact Journals • Effective Communication of Complex Data • Cross-functional Team Leadership • Interdisciplinary Teamwork & Leadership • Cross-Cultural & Multilingual Collaboration.

Professional Courses & Certificates

Foundations of Project Management 2025
Google Career Certificate, US

Bioinformatics Specialization 2025
University of California San Diego, US (Credential ID RLKNBVP83G66)

Finding Mutations in DNA and Proteins (Bioinformatics VI) 2025
University of California San Diego, US (Credential ID 9A23CVQVR253)

Principles of Management 2025
Saylor Academy (Credential ID 8949223071JO)

AI For Medical Treatment 2025
DeepLearning.AI (Credential ID O5JYW2R91P46)

AI for Medical Diagnosis 2025
DeepLearning.AI, US (Credential ID N3SY53I7VFPN)

AI for Medical Prognosis 2025
DeepLearning.AI (Credential ID BLAU4AME4ICT)

AI for Medicine Specialization 2025
DeepLearning.AI (Credential ID FSQ2LZ5PZZ73)
Algorithms for DNA Sequencing 2025
Johns Hopkins University, US (Credential ID 5P0J5NT6AQ7C)
Bioconductor for Genomic Data Science 2025
Johns Hopkins University, US (Credential ID PH8DN4TI4W39)
Bioinformatics Capstone: Big Data in Biology 2025
University of California San Diego, US (Credential ID 4N4E9HD6J064)
Bioinformatics and Drug Design 2025
IINS — International Institute of New Sciences, Iran (Credential ID 215172)
Command Line Tools for Genomic Data Science 2025
Johns Hopkins University, US (Credential ID X1MNAM87Y88G)
Comparing Genes, Proteins, and Genomes (Bioinformatics III) 2025
University of California San Diego, US (Credential ID 1KJMZON1CPQB)
Drug Delivery Systems Design 2025
IINS — International Institute of New Sciences, Iran (Credential ID 215171)
Finding Hidden Messages in DNA (Bioinformatics I) 2025
University of California San Diego, US (Credential ID XKAMNEN2117R)
Genome Sequencing (Bioinformatics II) 2025
University of California San Diego, US (Credential ID JY3T0C378FS2)
Genomic Data Science Specialization 2025
Johns Hopkins University, US (Credential ID P5KPHZAJRBHM)
Genomic Data Science and Clustering (Bioinformatics V) 2025
University of California San Diego, US (Credential ID DTJPNK9E6KWR)
ImmunoInformatics & Vaccine Design 2025
IINS — International Institute of New Sciences, Iran (Credential ID 215173)
Molecular Evolution (Bioinformatics IV) 2025
University of California San Diego, US (Credential ID SD7V7ECL5XH9)
Protein Design & Enzyme Engineering 2025
IINS — International Institute of New Sciences, Iran (Credential ID 215174)
Python for Genomic Data Science 2025
Johns Hopkins University, US (Credential ID 2ARU41Y2UOOZ)
Statistics for Genomic Data Science 2025
Johns Hopkins University, US (Credential ID SP9CG23IXOHO)
Drug Development 2025
University of California San Diego, US (Credential ID BCEA7GVHQOH5)
Drug Discovery 2025
University of California San Diego, US (Credential ID YKIXKWQQORE6)
Introduction to Genomic Technologies 2025
Johns Hopkins University, US (Credential ID MKRB8LVKXIV3)
Introduction to Medical Biochemistry 2025
Udemy, US (Credential ID UC-1e95d0a4-507b-49f3-be42-fa8aaa95a66a)
R Programming 2025
Johns Hopkins University, US (Credential ID Z4S9DX6LJ10E)
Convolutional Neural Networks 2024
DeepLearning.AI, US (Credential ID XWQYAPUY3BE5)
Neural Networks and Deep Learning 2024
DeepLearning.AI, US (Credential ID E0QCCZ6PUT4O)
Advanced Learning Algorithms 2024
DeepLearning.AI, Stanford University, US (Credential ID FV3POQPHC286)
Machine Learning Specialization 2024
DeepLearning.AI, Stanford University, US (Credential ID JHYA5KJI0ZYD)
Python for Data Science and Machine Learning Bootcamp 2024
Udemy, US (Credential ID UC-3d001b17-6ac7-4f3e-9f88-747814acaa16)
Supervised Machine Learning: Regression and Classification 2024
DeepLearning.AI, Stanford University, US (Credential ID 4OPDQSKOI9IR)
Unsupervised Learning, Recommenders, Reinforcement Learning 2024
DeepLearning.AI, Stanford University, US (Credential ID K6N6BJDCVINA)
Using Databases with Python 2024
University of Michigan, US (Credential ID F22M2AFDYNZH)

Using Python to Access Web Data 2024
University of Michigan, US (Credential ID U8A4PPFP7FEC)
Capstone: Retrieving, Processing, and Visualizing Data with Python 2024
University of Michigan, US (Credential ID WG5T4F6VLFQC)
Python Data Structures 2024
University of Michigan, US (Credential ID SAJDAZ26PVSY)
Data Analysis with Python 2024
Cognitive English Classes, India (Credential ID 40e38797349345398acc9e9561bfebbc)
Python for Data Science 2024
Cognitive English Classes, India (Credential ID 198fc11bb6f44c3f9abeead6db6f9b4c)
Advanced Mathematics 2012
Sharif University of Technology, Tehran, Iran

Workshop Participations

38th Annual Fred Scott Feline Symposium 2026
Cornell University College of Veterinary Medicine, Ithaca, NY, US
21st Northeastern Granular Materials Workshop 2025
Yale University, CT, US
18th Northeast Complex Fluids and Soft Matter Workshop 2023
Columbia University, NY, US
17th Northeast Complex Fluids and Soft Matter Workshop 2022
Stevens Institute of Technology, NJ, US

Professional Memberships

American Society of Human Genetics (ASHG) Jul 2025 – Now
American Institute of Chemical Engineers (AIChE) Sep 2024 – Sep 2025
American Physical Society (APS) Mar 2022 – Aug 2024

Honors & Awards

Journal Editor's Choice
Issued by Oxford University Press, UK Feb 2024
Article with DOI: 10.1093/ce/zkad085 selected as Editor's Choice, Oxford University Press
Columbia University Full Scholarship
Issued by Columbia University, New York City, NY, US Jan 2022
Full scholarship for PhD study and research at School of Engineering
DAAD Scholarship
Issued by German Academic Exchange Service, Germany Jan 2021
Full Scholarship (180k €) for Research at Technical University of Munich (TUM), Germany
Top Rank in Nationwide Entrance Examination for MS Studies
Issued by National Organization Educational Testing, Iran Sep 2012
Top 1% in Nationwide Entrance Examination for graduate studies (MS) among more than 3,100 participants
Top Rank in Nationwide Entrance Examination for BS Studies
Issued by National Organization Educational Testing, Iran Sep 2007
Top 1% in Nationwide Entrance Examination for undergraduate studies (BS), among more than 270,000 participants
4th-Rank Student in Diploma
Issued by Markazi Province Educational Office, Arak, Iran Sep 2006
4th-rank student in high school diploma at the city of Arak, Iran, among 1,177 students

Language Proficiency

Persian – Native Language
English – Full Professional Proficiency
French – Limited Working Proficiency (A2)
German – Limited Working Proficiency (A2)
Arabic – Limited Working Proficiency

References

Priscila Serpa, Assistant Professor, Cornell University, Ithaca, NY, US. Email: pserpa@cornell.edu
Christopher Boyce, Associate Professor, Columbia University, New York City, NY, US. Email: cmb2302@columbia.edu
Karim Mazaheri, Professor, Sharif University of Technology, Tehran, Iran. Email: mazaheri@sharif.edu
Niloufar Salimian, Assistant Professor, Azad University, Shahr-e-Kord, Iran. Email: n.salimian@iau.ir